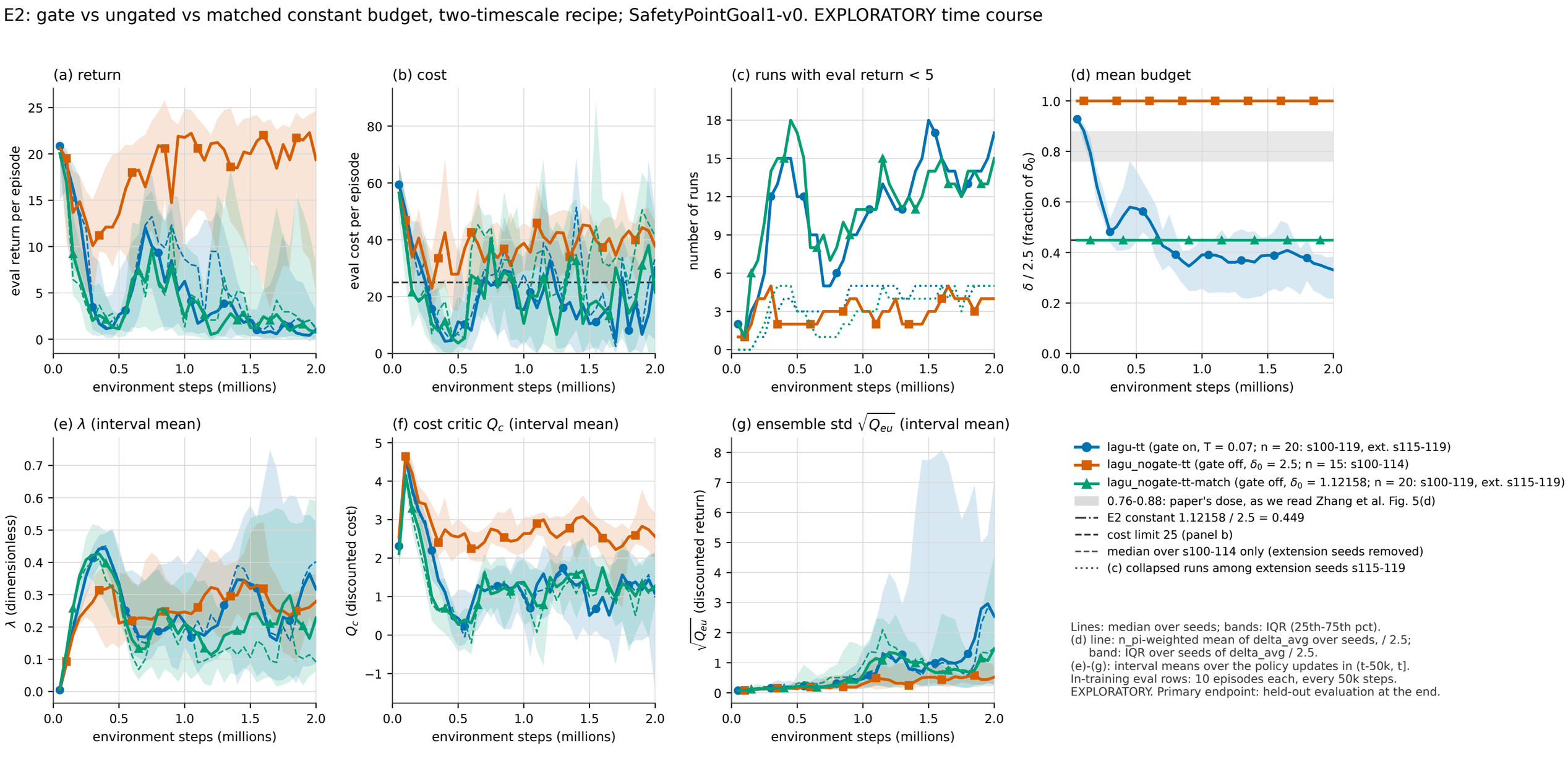
E2: gate vs ungated vs matched constant budget, two-timescale recipe; SafetyPointGoal1-v0. EXPLORATORY time course
(a) return
eval return per episode
environment steps (millions)
(b) cost
eval cost per episode
environment steps (millions)
(c) runs with eval return < 5
number of runs
environment steps (millions)
(d) mean budget
δ / 2.5 (fraction of δ0)
environment steps (millions)
(e) λ (interval mean)
λ (dimensionless)
environment steps (millions)
(f) cost critic Qc (interval mean)
Qc (discounted cost)
environment steps (millions)
(g) ensemble std √Qeu (interval mean)
√Qeu (discounted return)
environment steps (millions)
lagu-tt (gate on, T = 0.07; n = 20: s100-119, ext. s115-119)
lagu_nogate-tt (gate off, δ0 = 2.5; n = 15: s100-114)
lagu_nogate-tt-match (gate off, δ0 = 1.12158; n = 20: s100-119, ext. s115-119)
0.76-0.88: paper's dose, as we read Zhang et al. Fig. 5(d)
E2 constant 1.12158 / 2.5 = 0.449
cost limit 25 (panel b)
median over s100-114 only (extension seeds removed)
(c) collapsed runs among extension seeds s115-119
Lines: median over seeds; bands: IQR (25th-75th pct).
(d) line: n_pi-weighted mean of delta_avg over seeds, / 2.5;
band: IQR over seeds of delta_avg / 2.5.
(e)-(g): interval means over the policy updates in (t-50k, t].
In-training eval rows: 10 episodes each, every 50k steps.
EXPLORATORY. Primary endpoint: held-out evaluation at the end.